

Reference Pages

The Singleton Bound

If C is a binary code of length n and distance δ then

$$|C| \leq 2^{n+1-\delta}.$$

The Hamming Bound

If C is a binary code of length n and distance $\delta = 2t + 1$ or $\delta = 2t + 2$ then

$$|C| \leq \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{t}}.$$

The Griesmer Bound

Let C be a binary linear (n, k, δ) code. Then

$$n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil.$$

The Gilbert-Varshamov Bound

Let n, k and δ be positive integers with n and $\delta \geq 2$.

- If
$$\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2} < 2^{n-k}$$

then there exists a binary linear (n, k, δ) code.

- Further, there exists a binary linear code C with length n and distance δ with

$$|C| \geq \frac{2^{n-1}}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}.$$

Factorizations of $1 + x^n$ into irreducible polynomials

$$\begin{aligned} 1 + x &= 1 + x \\ 1 + x^3 &= (1 + x)(1 + x + x^2) \\ 1 + x^5 &= (1 + x)(1 + x + x^2 + x^3 + x^4) \\ 1 + x^7 &= (1 + x)(1 + x + x^3)(1 + x^2 + x^3) \\ 1 + x^9 &= (1 + x)(1 + x + x^2)(1 + x^3 + x^6) \\ 1 + x^{11} &= (1 + x)(1 + x + x^2 + \cdots + x^{10}) \\ 1 + x^{13} &= (1 + x)(1 + x + x^2 + \cdots + x^{12}) \\ 1 + x^{15} &= (1 + x)(1 + x + x^2)(1 + x + x^4)(1 + x^3 + x^4)(1 + x + x^2 + x^3 + x^4) \\ 1 + x^{17} &= (1 + x)(1 + x^3 + x^4 + x^5 + x^8)(1 + x + x^2 + x^4 + x^6 + x^7 + x^8) \\ 1 + x^{19} &= (1 + x)(1 + x + x^2 + \cdots + x^{18}) \\ 1 + x^{23} &= (1 + x)(1 + x + x^5 + x^6 + x^7 + x^9 + x^{11})(1 + x^2 + x^4 + x^5 + x^6 + x^{10} + x^{11}) \end{aligned}$$

Reference Pages (continued)

IMLD for binary cyclic linear codes & cyclic burst errors

To correct a cyclic burst error of length $\leq t$ using parity-check matrix

$\mathbf{H}$:

$$\mathbf{H} = \begin{pmatrix} r_{n-k} \\ r_{n-k+1} \\ r_{n-k+2} \\ \vdots \\ r_{n-1} \\ \mathbf{I}_{n-k} \end{pmatrix}$$

- Receive w . Calculate the syndrome s_0 of w .
- Calculate $s_i(x) = xs_{i-1}(x) \bmod g(x)$ until an s_i is found with a burst error of length at most t .
- Then $e = x^{k-i}s_i(x) = x^{n+k-i}s_i(x) \bmod x^n + 1$.

IMLD for the binary Extended Golay Code

C_{24}

The Golay Code C_{23} & The Extended Golay Code C_{24}

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}$$

- Let θ_i be the length 12 weight one word with a 1 in position i .
- Receive w . Compute the syndrome $s = w\mathbf{H} = w \begin{pmatrix} \mathbf{I}_{12} \\ \mathbf{B} \end{pmatrix}$.
- If $\|s\| \leq 3$ then $e = (s \mid 0)$.
- If $\|s + b_i\| \leq 2$ for some row b_i of $\mathbf{B}$ then $e = (s + b_i \mid \theta_i)$.
- Compute the second syndrome $s\mathbf{B}$.
- If $\|s\mathbf{B}\| \leq 3$ then $e = (0 \mid s\mathbf{B})$.
- If $\|s\mathbf{B} + b_i\| \leq 2$ for some row b_i of $\mathbf{B}$ then $e = (\theta_i \mid s\mathbf{B} + b_i)$.
- If e is not yet determined, more than 3 errors have occurred so request retransmission.

IMLD for the binary Golay Code C_{23}

- Form $w^* = w0$ or $w^* = w1$ so that w^* has odd weight.
- Decode w^* to a codeword $c^* \in C_{24}$ using the Algorithm for IMLD of C_{24} .
- Remove the last bit from c^* to obtain a codeword $c \in C_{23}$.

English letter equivalents in $\mathbb{Z}_{26}$

a,A	b,B	c,C	d,D	e,E	f,F	g,G	h,H	i,I	j,J	k,K	l,L	m,M
0	1	2	3	4	5	6	7	8	9	10	11	12
n,N	o,O	p,P	q,Q	r,R	s,S	t,T	u,U	v,V	w,W	x,X	y,Y	z,Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Correspondence between bistrings and hexadecimal digits

0000	0001	0010	0011	0100	0101	0110	0111
0	1	2	3	4	5	6	7
1000	1001	1010	1011	1100	1101	1110	1111
8	9	a	b	c	d	e	f